



Engineering Electromagnetics

工程电磁场

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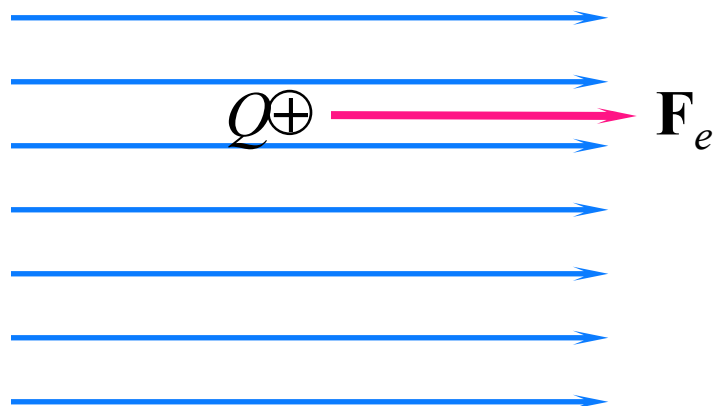
11: The Steady Magnetic Field



- Force on a Moving Charge

The forces exerted on a point charge by electric and magnetic fields are:

charge can be stationary
or moving

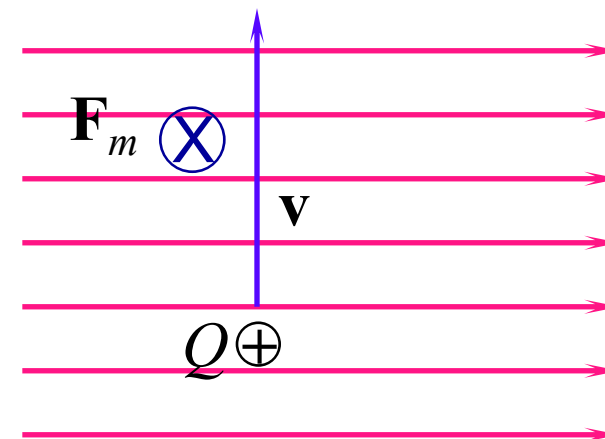


E

$$\mathbf{F}_e = Q\mathbf{E}$$

in the direction of **E**

charge is moving
at velocity **v**



B

$$\mathbf{F}_m = Q (\mathbf{v} \times \mathbf{B})$$

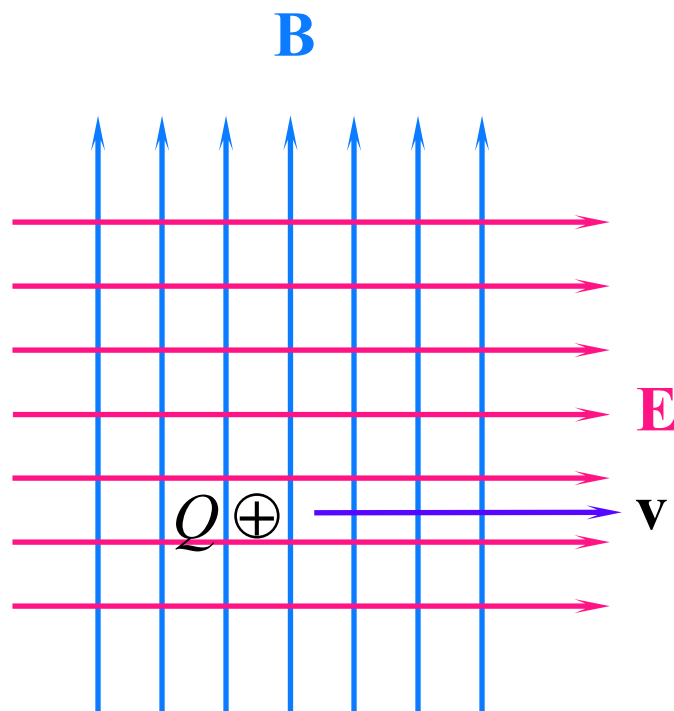
into the screen

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- Lorentz Force Law

Generally, with both electric and magnetic fields present, we have both forces:



The electric field will, in this case, accelerate the charge in the direction of $\mathbf{E}$, making it cross the $\mathbf{B}$ field lines in the perpendicular sense; this gives a magnetic force component that is out of the screen

The total force on the moving charge is then the sum of the two, or

$$\mathbf{F} = Q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$$

This is the *Lorentz Force Law* (sometimes called the “**fifth Maxwell equation**”)

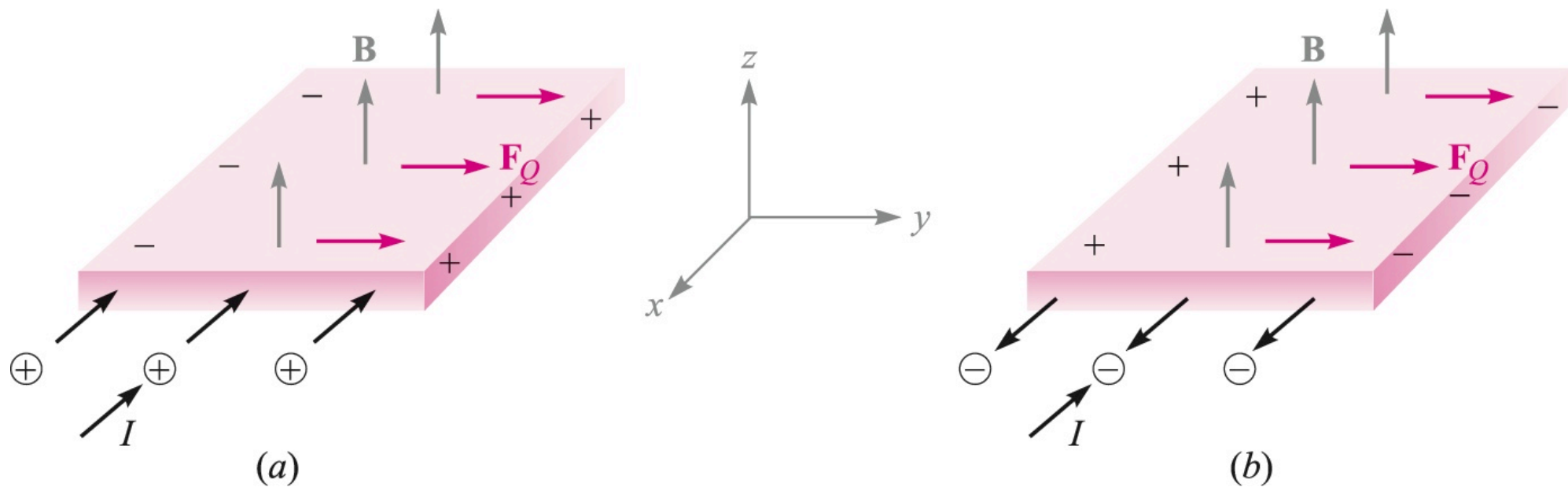
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- Hall Effect

When a $\mathbf{B}$ field is applied in a direction perpendicular to a current, positive and negative carriers will be displaced slightly, as shown, as a result of the magnetic forces on the moving charges.

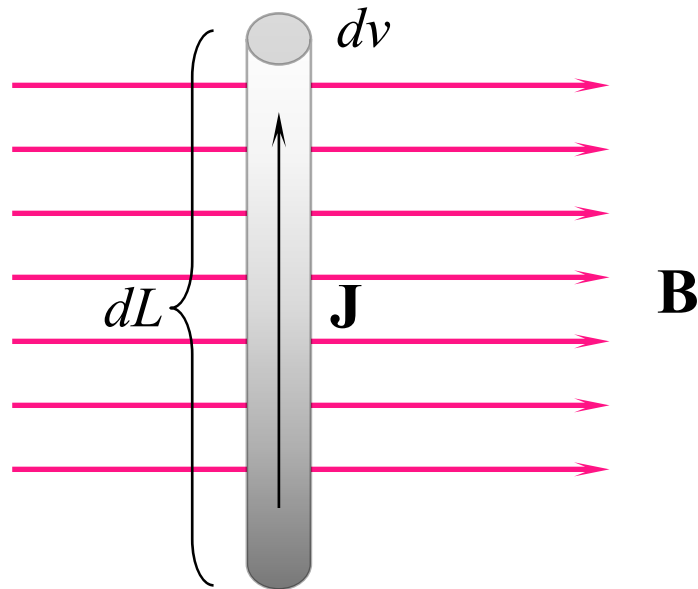
This produces a measurable voltage, known as the *Hall Voltage*.



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- Force on a **Differential Current Element**



Consider a small segment (length dL) of current in the form of a **volume current density $\mathbf{J}$** , suspended in a **magnetic field, $\mathbf{B}$** . The current element has volume dv .

We know that current density is volume charge density moving at velocity $\mathbf{v}$:

$$\mathbf{J} = \rho_v \mathbf{v}$$

..and we can write **the differential force on a differential charge, dQ** :

$$d\mathbf{F} = dQ \mathbf{v} \times \mathbf{B} \quad \text{where} \quad dQ = \rho_v dv$$

Therefore: $d\mathbf{F} = \rho_v dv \mathbf{v} \times \mathbf{B}$..so that finally: $d\mathbf{F} = \mathbf{J} \times \mathbf{B} dv$

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- Other Expressions for Differential Force

For volume current, surface current, or filament current, we have the appropriate expressions for differential current:

$$\mathbf{J} d\mathbf{v} = \mathbf{K} d\mathbf{S} = I d\mathbf{L}$$

The corresponding expressions for differential force within magnetic field $\mathbf{B}$ are:

$$d\mathbf{F} = \mathbf{J} \times \mathbf{B} d\mathbf{v}$$

volume current density (three dimensions)

$$d\mathbf{F} = \mathbf{K} \times \mathbf{B} d\mathbf{S}$$

surface current density (two dimensions)

$$d\mathbf{F} = I d\mathbf{L} \times \mathbf{B}$$

filament current of length dL (one dimension)

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- Evaluating the Total Force

In three or two dimensions,
the net force is found by
integrating over the volume
or surface that the current occupies

$$\left\{ \begin{array}{l} \mathbf{F} = \int_{\text{vol}} \mathbf{J} \times \mathbf{B} d\mathbf{v} \\ \mathbf{F} = \int_S \mathbf{K} \times \mathbf{B} dS \end{array} \right.$$

For a filament current, the total force will be the integral of the differential force, generally taken over the closed path that comprises the current:

$$\mathbf{F} = \oint I d\mathbf{L} \times \mathbf{B} = -I \oint \mathbf{B} \times d\mathbf{L}$$

For a straight filament of length L , having uniform current, and within a uniform field, this becomes:

$$\mathbf{F} = I\mathbf{L} \times \mathbf{B}$$

which in turn reduces to $F = BIL \sin \theta$

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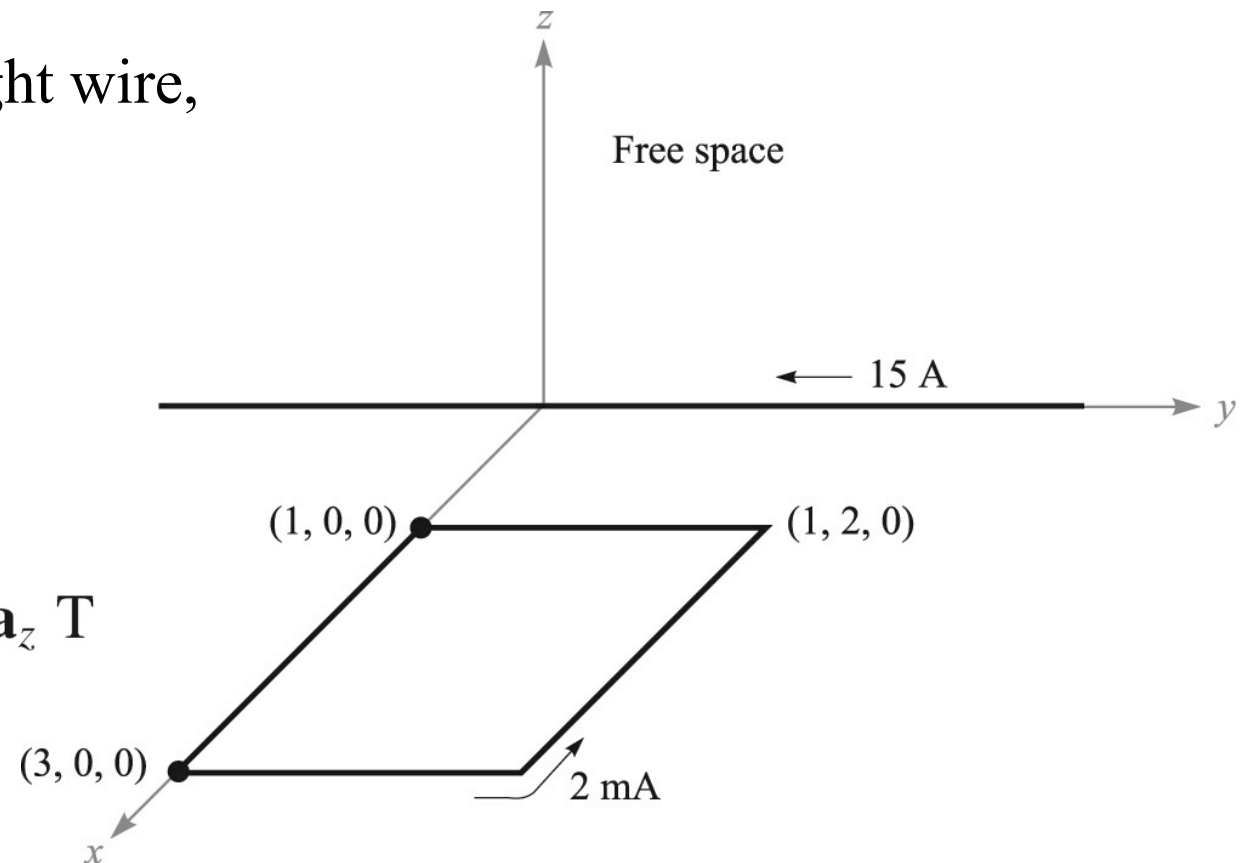
- Example: Force on a Square Current Loop

The magnetic field arising from the straight wire, evaluated in the plane of the loop, is:

$$\mathbf{H} = \frac{I}{2\pi x} \mathbf{a}_z = \frac{15}{2\pi x} \mathbf{a}_z \text{ A/m}$$

The $\mathbf{B}$ field is then:

$$\mathbf{B} = \mu_0 \mathbf{H} = 4\pi \times 10^{-7} \mathbf{H} = \frac{3 \times 10^{-6}}{x} \mathbf{a}_z \text{ T}$$



The total force on the loop is then found by evaluating:

$$\mathbf{F} = -I \oint \mathbf{B} \times d\mathbf{L}$$

over all four loop segments

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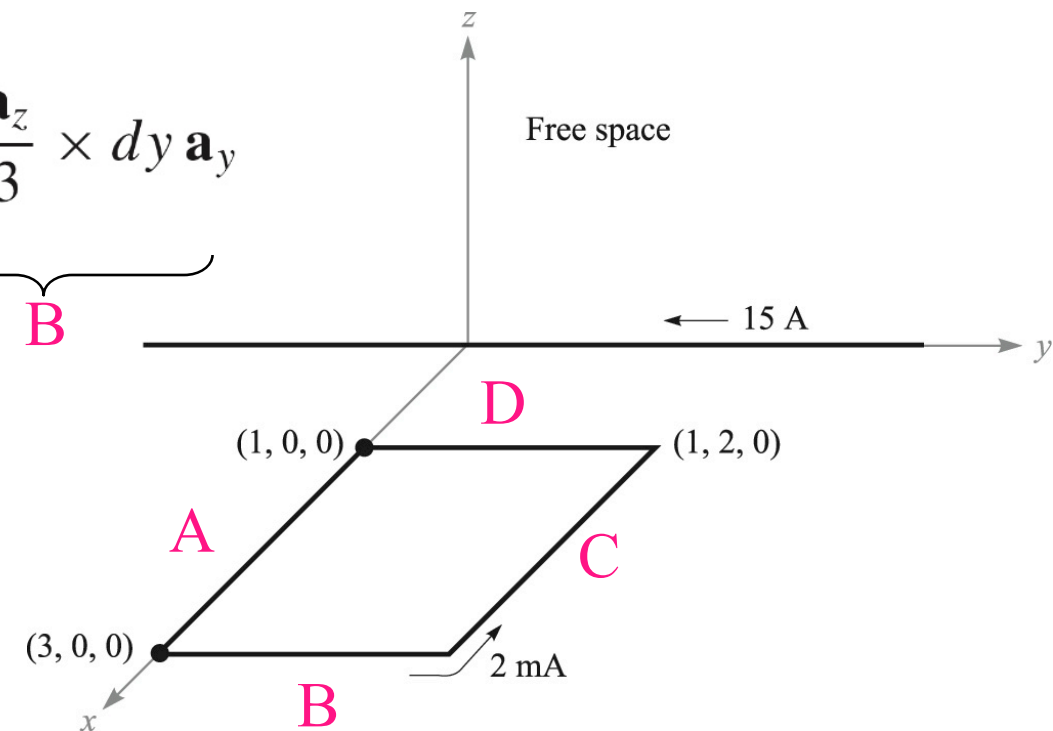


- Example: Force on a Square Current Loop

The integral becomes:

$$\mathbf{F} = -2 \times 10^{-3} \times 3 \times 10^{-6} \left[\underbrace{\int_{x=1}^3 \frac{\mathbf{a}_z}{x} \times dx \mathbf{a}_x}_A + \underbrace{\int_{y=0}^2 \frac{\mathbf{a}_z}{3} \times dy \mathbf{a}_y}_B \right]$$

$$+ \underbrace{\int_{x=3}^1 \frac{\mathbf{a}_z}{x} \times dx \mathbf{a}_x}_C + \underbrace{\int_{y=2}^0 \frac{\mathbf{a}_z}{1} \times dy \mathbf{a}_y}_D$$



$$= -6 \times 10^{-9} \left[\ln x \Big|_1^3 \mathbf{a}_y + \frac{1}{3} y \Big|_0^2 (-\mathbf{a}_x) + \ln x \Big|_3^1 \mathbf{a}_y + y \Big|_2^0 (-\mathbf{a}_x) \right]$$

$$= -6 \times 10^{-9} \left[(\ln 3) \mathbf{a}_y - \frac{2}{3} \mathbf{a}_x + \left(\ln \frac{1}{3} \right) \mathbf{a}_y + 2 \mathbf{a}_x \right]$$

$$= -8 \mathbf{a}_x \text{ nN}$$

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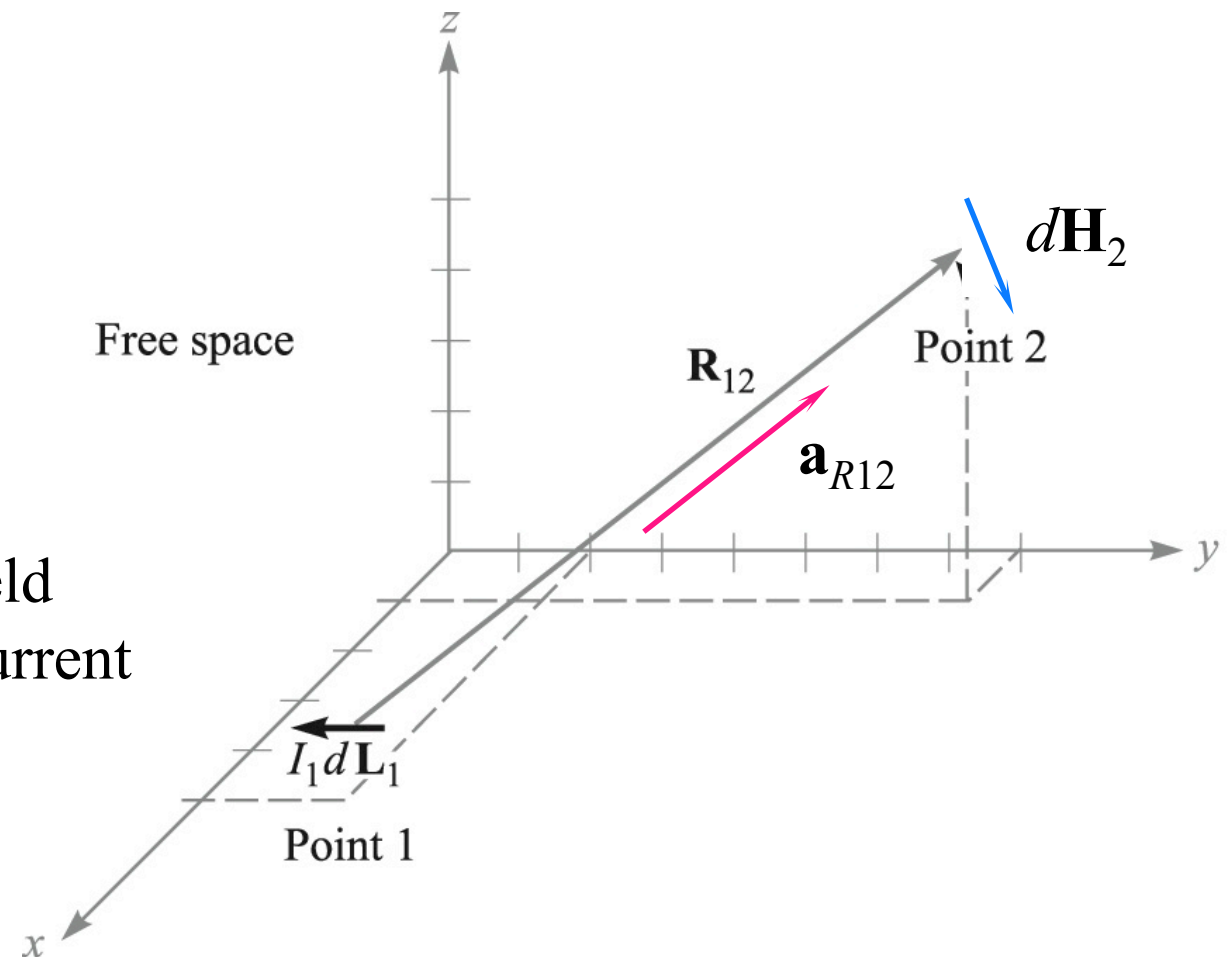


- Force Between Differential Current Elements

We use the **Biot-Savart Law** to find the **differential magnetic field at Point 2** that arises from the differential current element at Point 1:

$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$

The flux density associated with this field will generate a force on an additional current element at Point 2



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- Force Between Differential Current Elements

A second differential current element is now placed at Point 2. It will experience a force given by:

$$d(d\mathbf{F}_2) = I_2 d\mathbf{L}_2 \times d\mathbf{B}_2$$

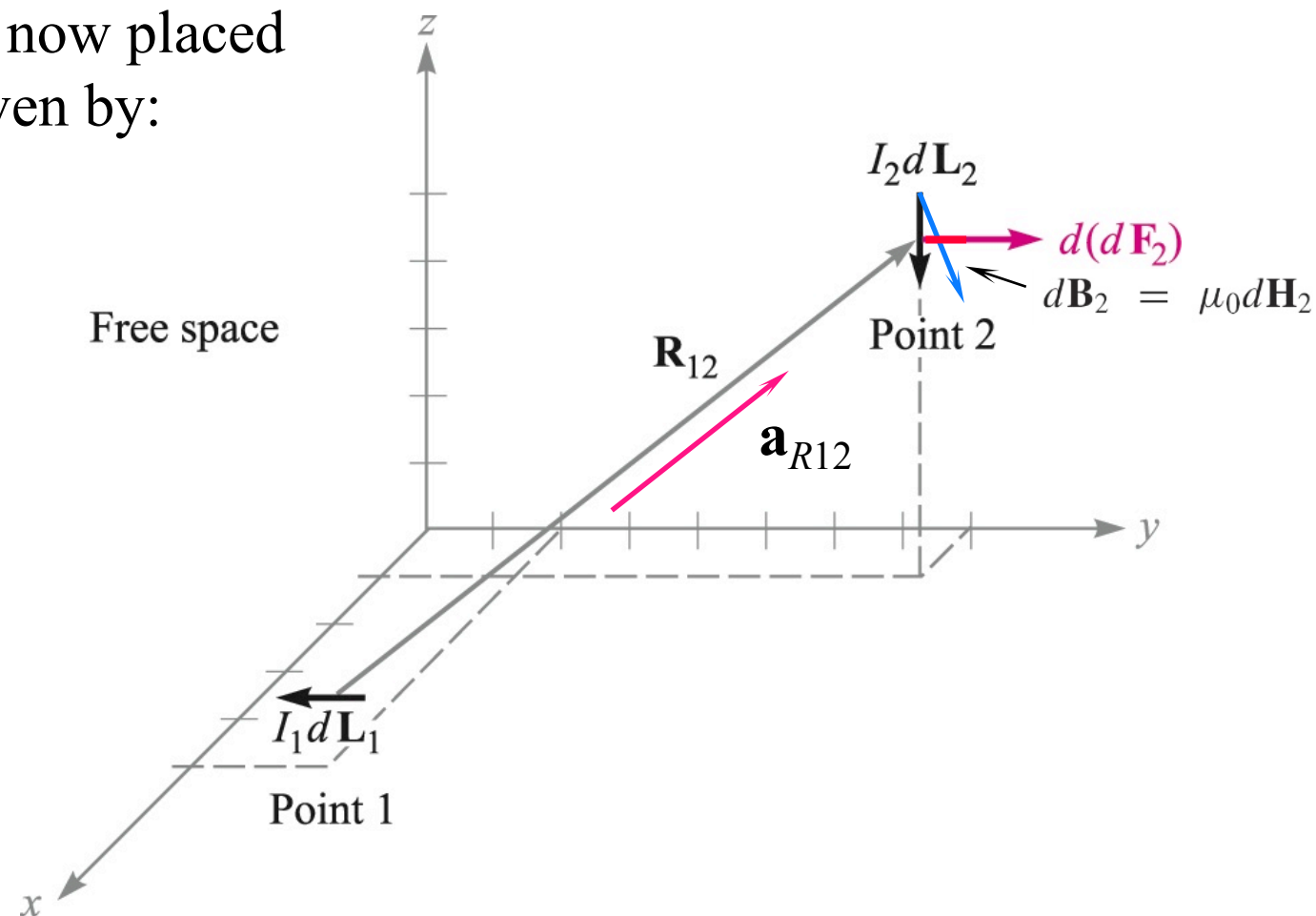
Then with:

$$d\mathbf{H}_2 = \frac{I_1 d\mathbf{L}_1 \times \mathbf{a}_{R12}}{4\pi R_{12}^2}$$

where: $d\mathbf{B}_2 = \mu_0 d\mathbf{H}_2$

we finally obtain:

$$d(d\mathbf{F}_2) = \mu_0 \frac{I_1 I_2}{4\pi R_{12}^2} d\mathbf{L}_2 \times (d\mathbf{L}_1 \times \mathbf{a}_{R12})$$



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- Example

Given: $I_1 d\mathbf{L}_1 = -3\mathbf{a}_y \text{ A} \cdot \text{m}$ at $P_1(5, 2, 1)$

$I_2 d\mathbf{L}_2 = -4\mathbf{a}_z \text{ A} \cdot \text{m}$ at $P_2(1, 8, 5)$

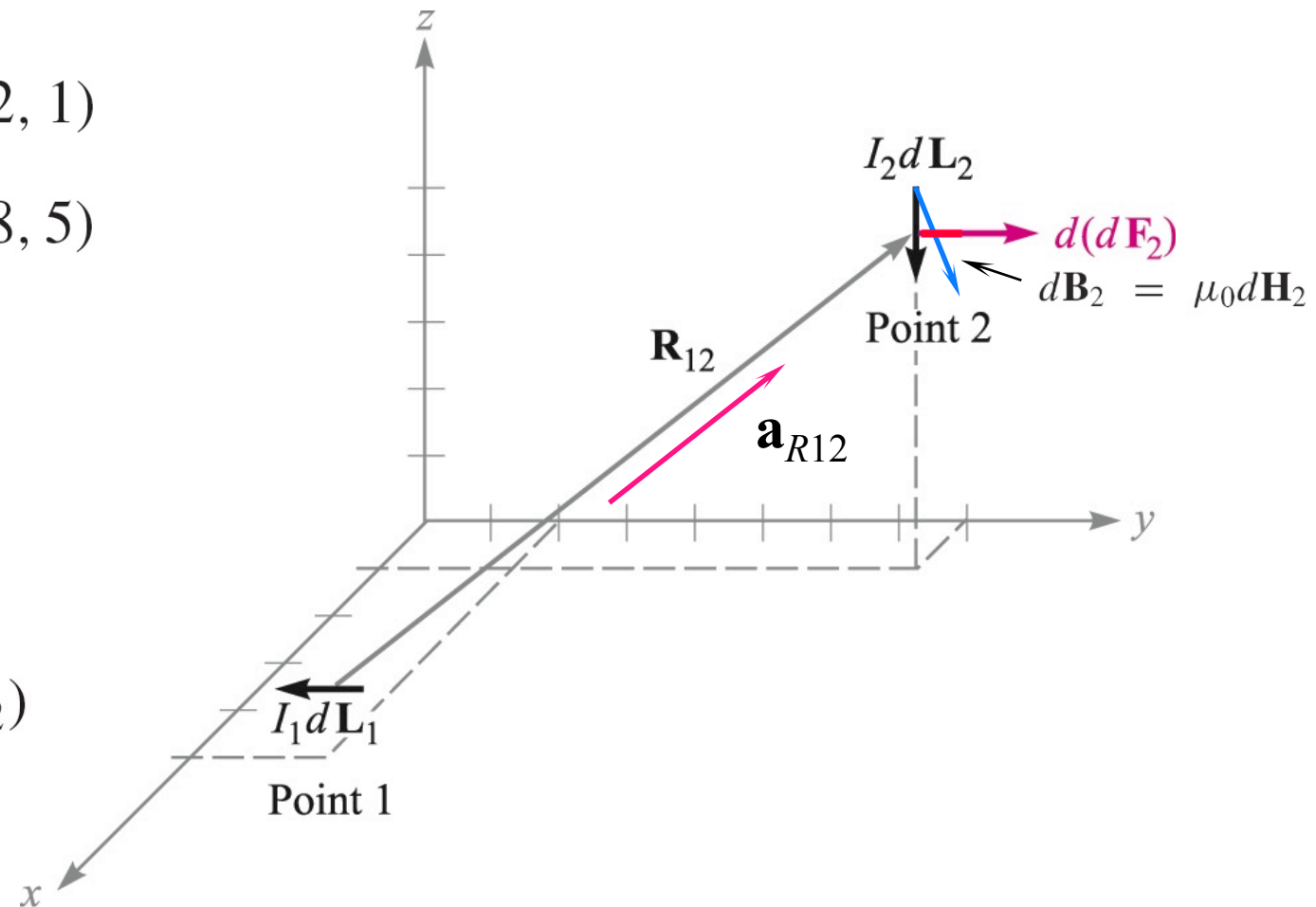
Then $\mathbf{R}_{12} = -4\mathbf{a}_x + 6\mathbf{a}_y + 4\mathbf{a}_z$

Substitute these into:

$$d(d\mathbf{F}_2) = \mu_0 \frac{I_1 I_2}{4\pi R_{12}^2} d\mathbf{L}_2 \times (d\mathbf{L}_1 \times \mathbf{a}_{R12})$$

Obtain:

$$\begin{aligned} d(d\mathbf{F}_2) &= \frac{4\pi 10^{-7}}{4\pi} \frac{(-4\mathbf{a}_z) \times [(-3\mathbf{a}_y) \times (-4\mathbf{a}_x + 6\mathbf{a}_y + 4\mathbf{a}_z)]}{(16 + 36 + 16)^{1.5}} \\ &= \underline{8.56\mathbf{a}_y \text{ nN}} \end{aligned}$$



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- Force Between Current Filaments of Finite or Infinite Lengths

The force is found by integrating the differential filament result:

$$d(d\mathbf{F}_2) = \mu_0 \frac{I_1 I_2}{4\pi R_{12}^2} d\mathbf{L}_2 \times (d\mathbf{L}_1 \times \mathbf{a}_{R12})$$

in which **the integral is taken over the lengths of both filaments**. To be complete, both integrals would be taken over the closed loops that the currents must form, thus the general expression:

$$\begin{aligned}\mathbf{F}_2 &= \mu_0 \frac{I_1 I_2}{4\pi} \oint \left[d\mathbf{L}_2 \times \oint \frac{d\mathbf{L}_1 \times \mathbf{a}_{R12}}{R_{12}^2} \right] \\ &= \mu_0 \frac{I_1 I_2}{4\pi} \oint \left[\oint \frac{\mathbf{a}_{R12} \times d\mathbf{L}_1}{R_{12}^2} \right] \times d\mathbf{L}_2\end{aligned}$$

These integrals can be **modified to incorporate specific limits**, in order to find the force from a **specified segment** of a wire that acts on a **specific segment** of another wire

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- Example: Force Between Parallel Wires (the easy way)

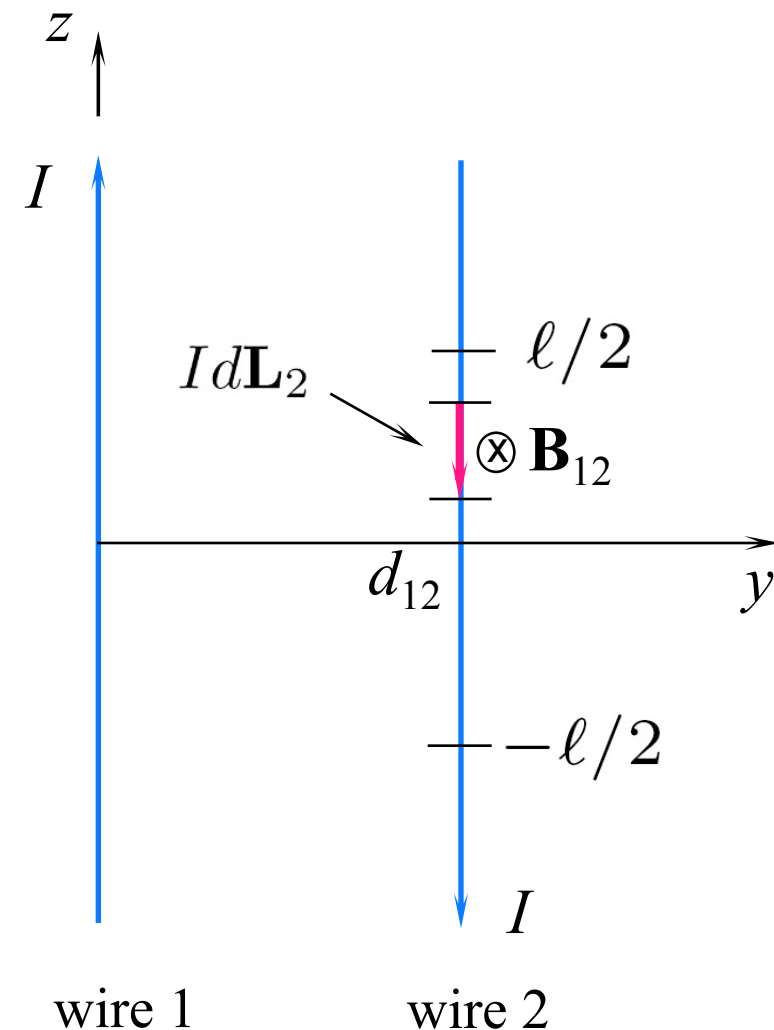
Consider the two wires shown here, **carrying equal and opposite currents**, I , and spaced by distance d_{12} along the y axis. The wires are oriented in the z direction and both are **infinitely long**.

The force is to be found on a length ℓ of wire 2, of equal extent above and below the y axis.

We can solve this one fairly quickly by observing that the $\mathbf{B}$ field from wire 1 at the location of wire 2 will be:

$$\mathbf{B}_{12} = -\frac{\mu_0 I}{2\pi d_{12}} \mathbf{a}_x$$

and $I d\mathbf{L}_2 = -I dz_2 \mathbf{a}_z$



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- Example: Force Between Parallel Wires (the easy way)

We have: $\mathbf{B}_{12} = -\frac{\mu_0 I}{2\pi d_{12}} \mathbf{a}_x$ and $I d\mathbf{L}_2 = -I dz_2 \mathbf{a}_z$

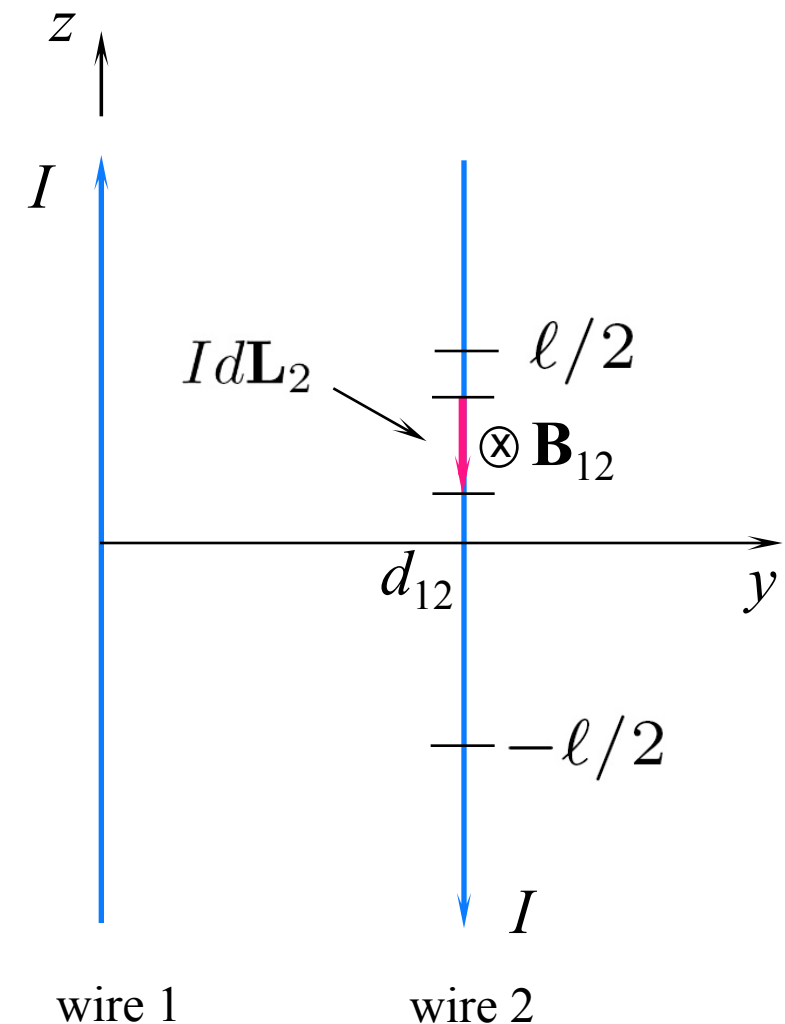
Then the force acting on the differential element shown is:

$$\begin{aligned} d\mathbf{F}_2 &= I d\mathbf{L}_2 \times \mathbf{B}_{12} \\ &= -I dz_2 \mathbf{a}_z \times \left(-\frac{\mu_0 I}{2\pi d_{12}} \mathbf{a}_x \right) = \frac{\mu_0 I^2 dz_2}{2\pi d_{12}} \mathbf{a}_y \end{aligned}$$

The total force on the length is therefore:

$$\mathbf{F}_2 = \int_{-\ell/2}^{\ell/2} I d\mathbf{L}_2 \times \mathbf{B}_{12} = \frac{\mu_0 I^2 \ell}{2\pi d_{12}} \mathbf{a}_y$$

....which is a repulsive force

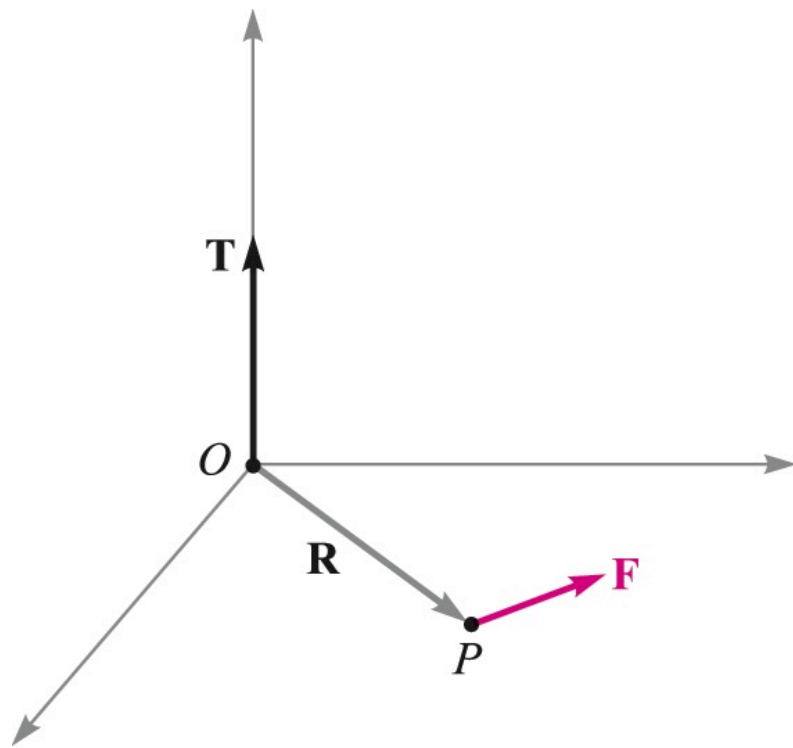


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- Torque 转矩: Basic Definition

Given a force $\mathbf{F}$ at point P , the **torque about the origin** is a vector that **is perpendicular to** the plane containing $\mathbf{F}$ and position vector $\mathbf{R}$. The **torque vector** is the cross product:



$$\underline{\mathbf{T} = \mathbf{R} \times \mathbf{F}}$$

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- Torque: General Equation

Now, consider two *equal and opposite* forces, applied at points P_1 and P_2 as shown.

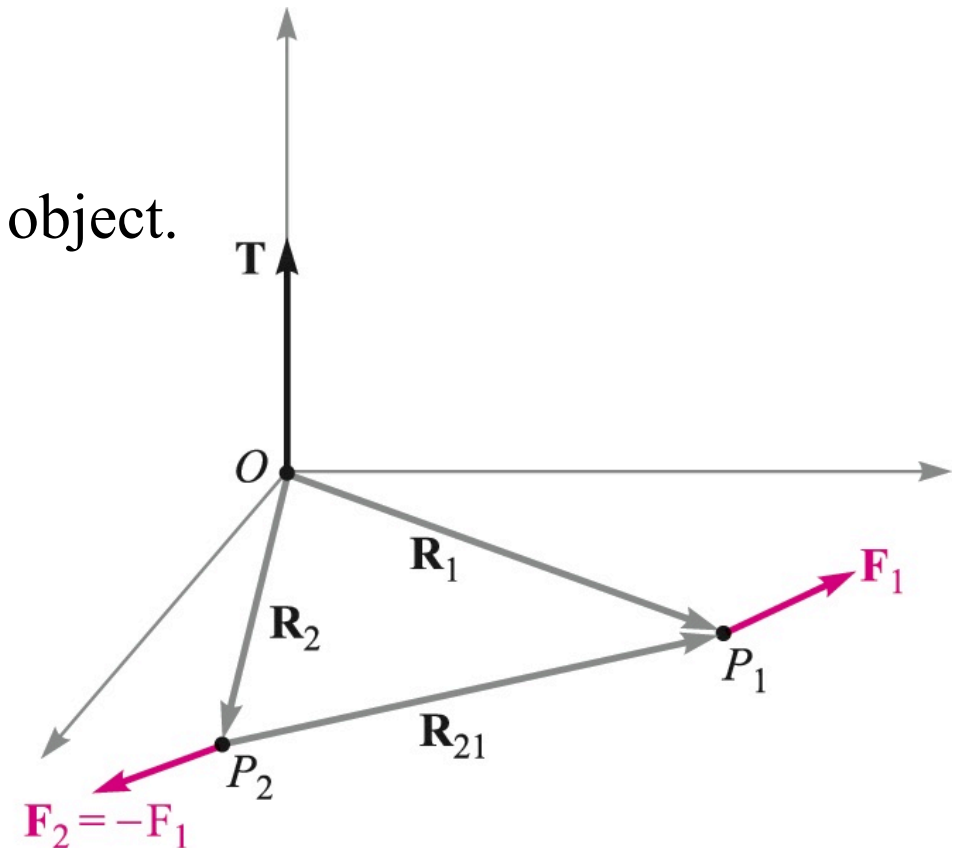
As the *net force* is zero, there is no translation of the object.

The *net torque* on the system will be:

$$\mathbf{T} = \mathbf{R}_1 \times \mathbf{F}_1 + \mathbf{R}_2 \times \mathbf{F}_2$$

But since $\mathbf{F}_2 = -\mathbf{F}_1$:

$$\mathbf{T} = (\mathbf{R}_1 - \mathbf{R}_2) \times \mathbf{F}_1 = \underline{\mathbf{R}_{21} \times \mathbf{F}_1}$$



This means that *the torque is independent of the choice of origin, provided the total force acting is zero*. This applies to *any number of forces*. And the origin can be located anywhere that is convenient.

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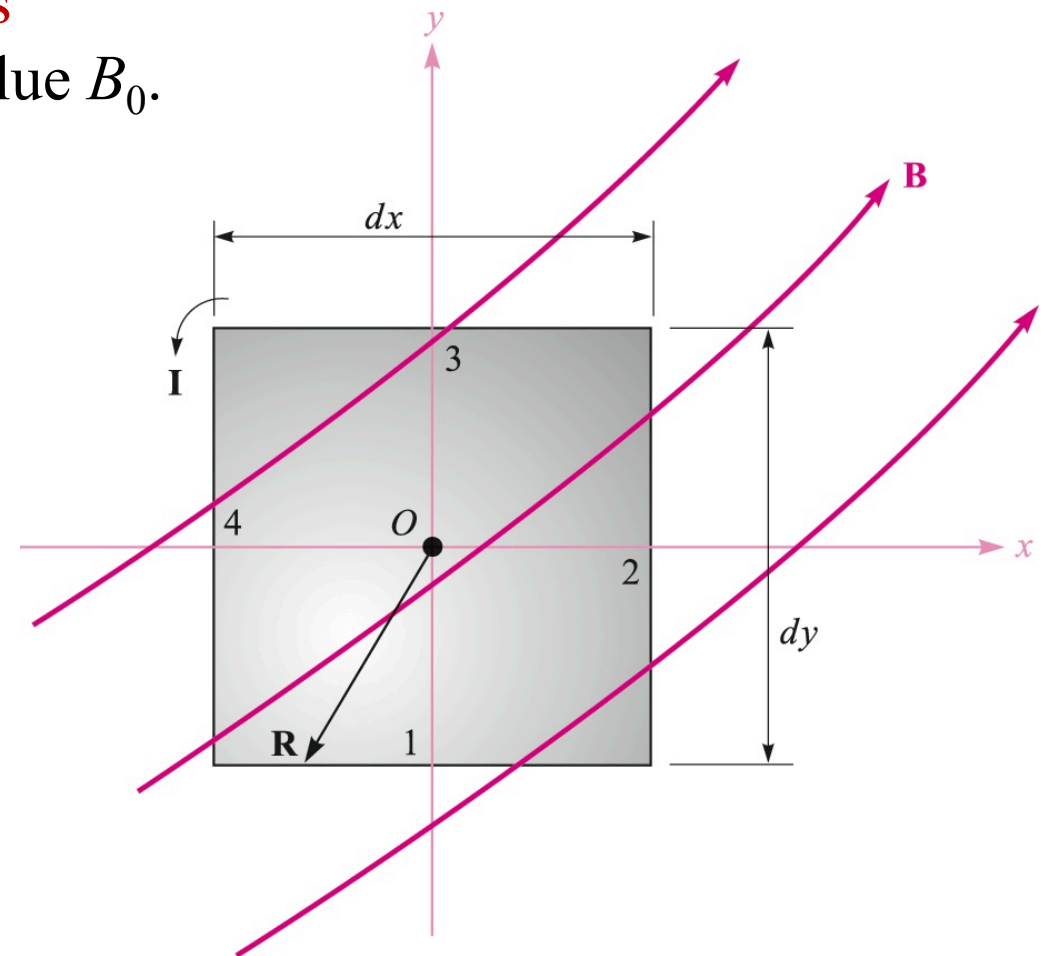


- Torque on a **Differential Current Loop**

The **filament loop** shown here lies in the xy plane with its center at the origin. Magnetic flux density $\mathbf{B}$ exists everywhere, and in a general direction. **As the loop is of differential size, the magnitude of $\mathbf{B}$ is assumed uniform over the loop area**, and has value B_0 . Current I circulates around the loop.

The differential force acting on side 1 is:

$$\begin{aligned} d\mathbf{F}_1 &= I dx \mathbf{a}_x \times \mathbf{B}_0 \\ &= I dx (B_{0y} \mathbf{a}_z - B_{0z} \mathbf{a}_y) \end{aligned}$$



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- Torque on a **Differential Current Loop**

We have the differential force acting on side 1:

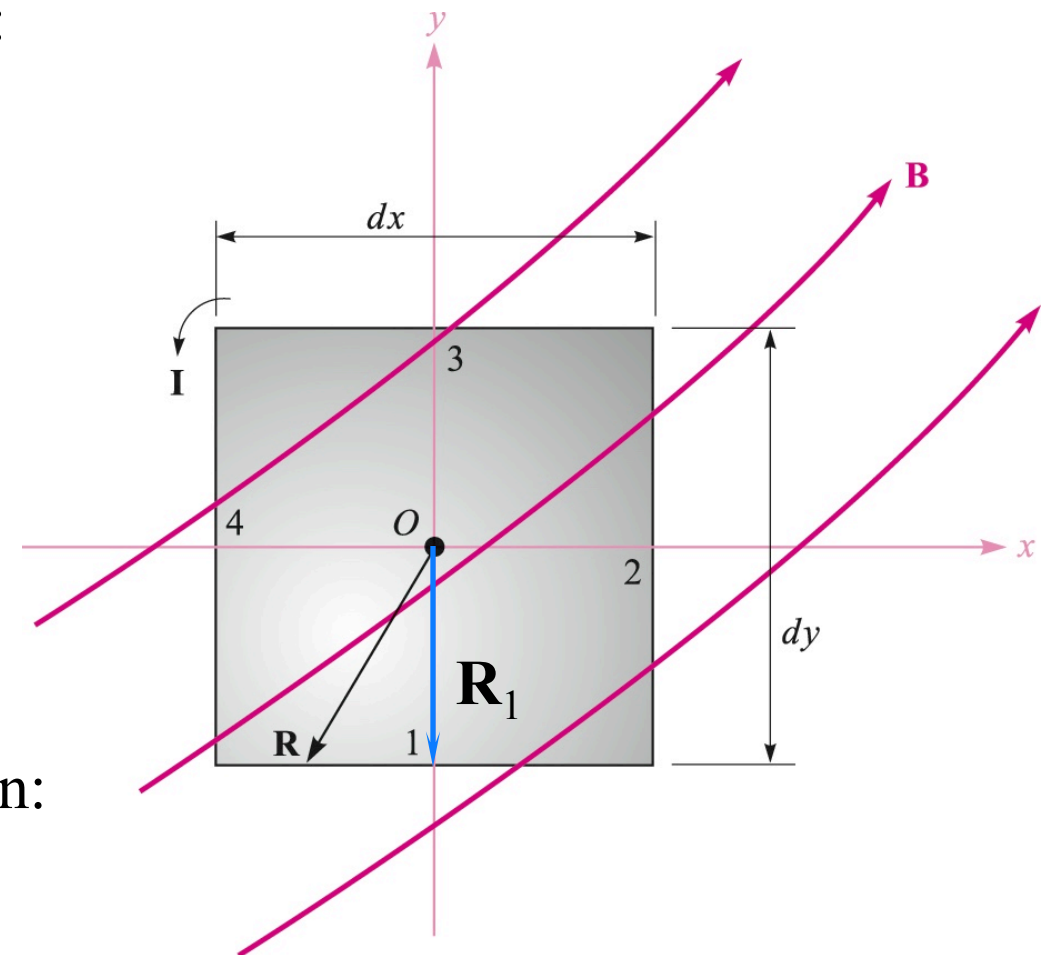
$$d\mathbf{F}_1 = I dx (B_{0y} \mathbf{a}_z - B_{0z} \mathbf{a}_y)$$

For side 1, the **lever arm** extends from the origin to the midpoint of the side, and is given by:

$$\mathbf{R}_1 = -\frac{1}{2} dy \mathbf{a}_y$$

The differential torque acting on side 1 is then:

$$\begin{aligned} d\mathbf{T}_1 &= \mathbf{R}_1 \times d\mathbf{F}_1 \\ &= -\frac{1}{2} dy \mathbf{a}_y \times I dx (B_{0y} \mathbf{a}_z - B_{0z} \mathbf{a}_y) \\ &= -\frac{1}{2} dx dy I B_{0y} \mathbf{a}_x \end{aligned}$$



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- Torque on a **Differential Current Loop**

We next consider the opposite side (3), and using similar reasoning, find the differential torque acting on side 3:

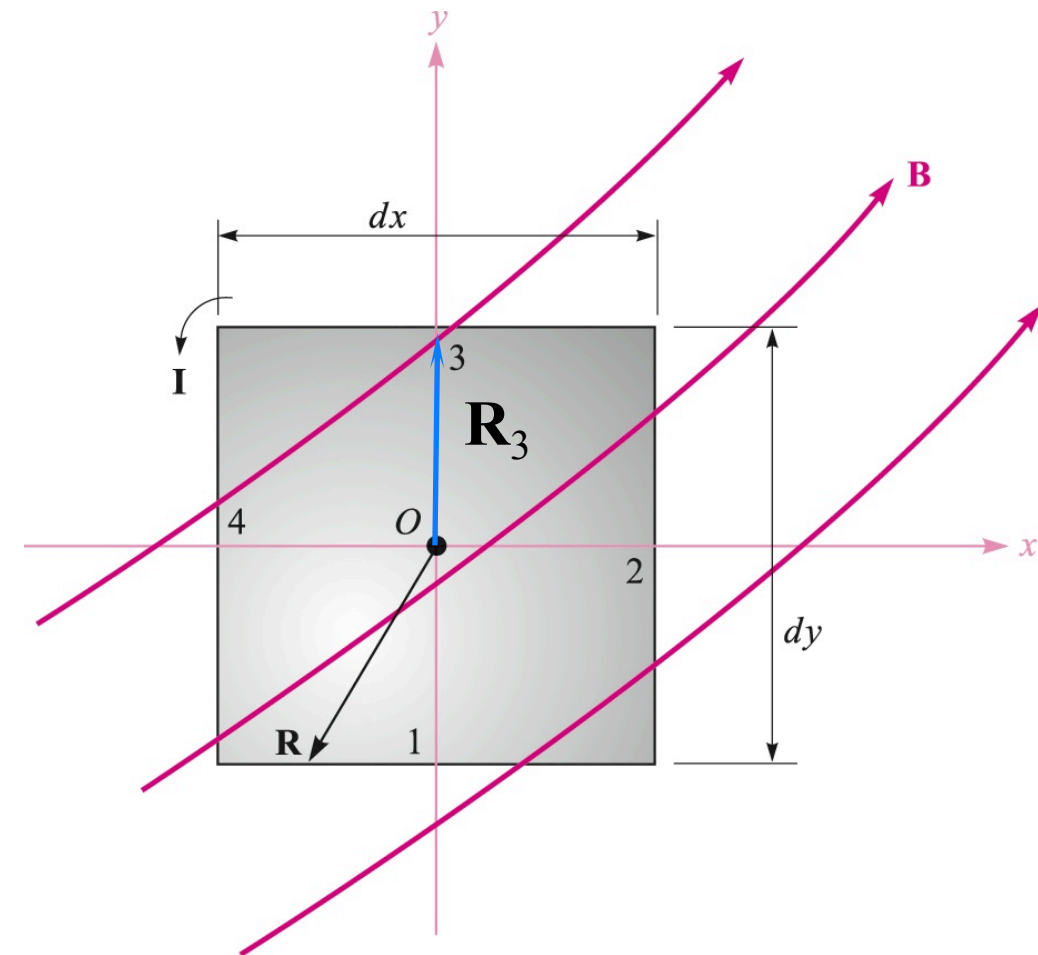
$$\begin{aligned} d\mathbf{T}_3 &= \mathbf{R}_3 \times d\mathbf{F}_3 \\ &= \frac{1}{2}dy \mathbf{a}_y \times (-I dx \mathbf{a}_x \times \mathbf{B}_0) \\ &= -\frac{1}{2}dx dy IB_{0y} \mathbf{a}_x = d\mathbf{T}_1 \quad \star \end{aligned}$$

The total differential torque acting on sides 1 and 3 is then:

$$d\mathbf{T}_1 + d\mathbf{T}_3 = -dx dy IB_{0y} \mathbf{a}_x$$

Then, using the same reasoning, the total differential torque acting on sides 2 and 4 is:

$$d\mathbf{T}_2 + d\mathbf{T}_4 = dx dy IB_{0x} \mathbf{a}_y$$



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- Torque on a **Differential Current Loop**

Now, with: $d\mathbf{T}_1 + d\mathbf{T}_3 = -dx dy IB_{0y} \mathbf{a}_x$

and $d\mathbf{T}_2 + d\mathbf{T}_4 = dx dy IB_{0x} \mathbf{a}_y$

The total torque on all four sides is:

$$d\mathbf{T} = I dx dy (B_{0x} \mathbf{a}_y - B_{0y} \mathbf{a}_x)$$

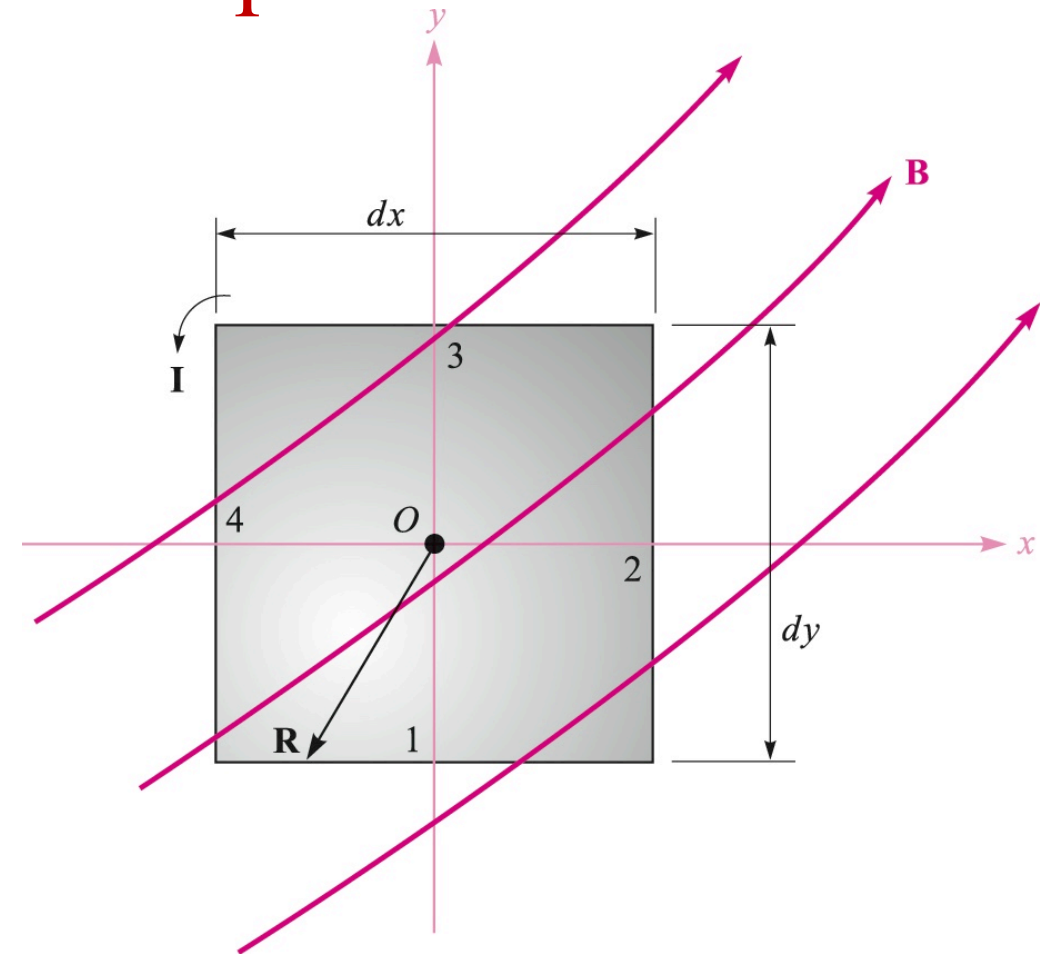
The terms in parenthesis can be written as a cross product:

$$d\mathbf{T} = I dx dy (\mathbf{a}_z \times \mathbf{B}_0)$$

Finally resulting in:

$$d\mathbf{T} = I d\mathbf{S} \times \mathbf{B}$$

$d\mathbf{S}$ points out of the screen in this example



where the **differential loop area vector** is defined using the right hand convention:

$d\mathbf{S} = dx dy \mathbf{a}_z$ Fingers in direction of current, thumb in direction of $\mathbf{S}$

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- Differential Magnetic Moment

Having found the torque on a differential current loop:

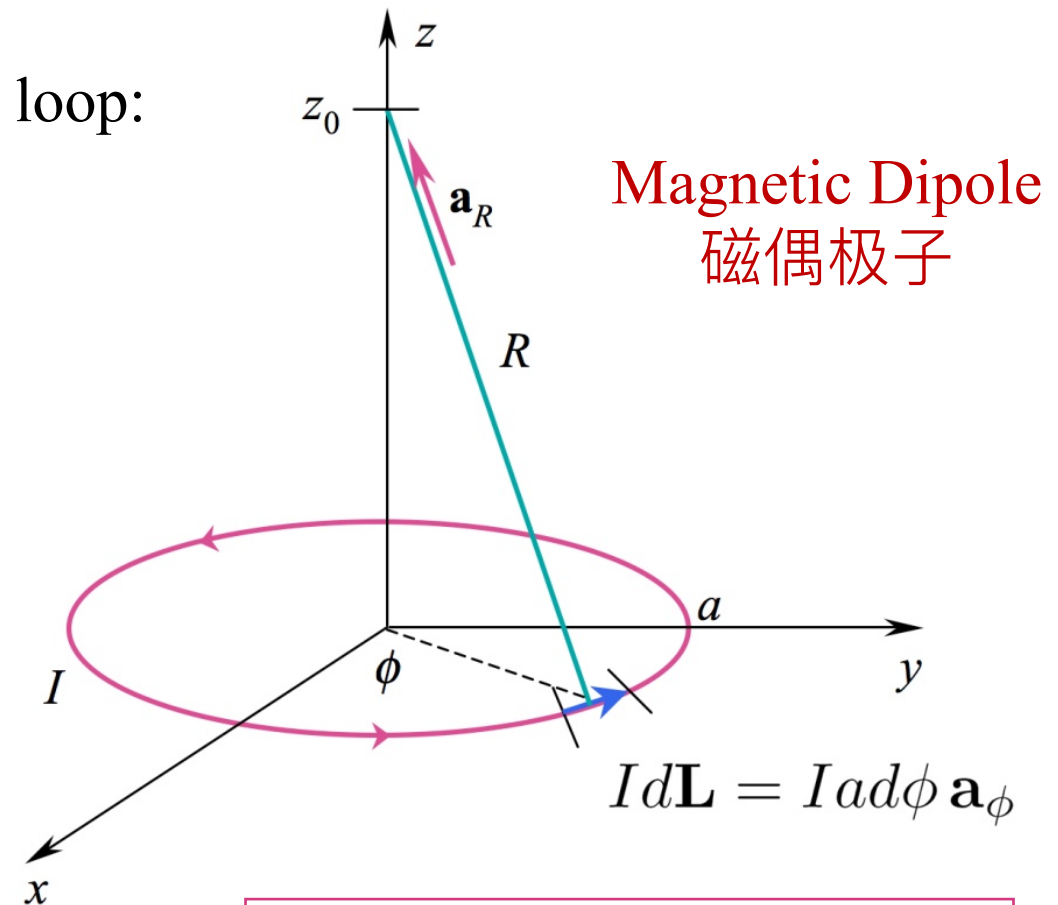
$$d\mathbf{T} = I d\mathbf{S} \times \mathbf{B}$$

Define the *differential magnetic dipole moment* as the product of the current and the differential area vector:

$$d\mathbf{m} = I d\mathbf{S}$$

from which:

$$d\mathbf{T} = d\mathbf{m} \times \mathbf{B}$$



$$\mathbf{H} = \frac{I(\pi a^2) \mathbf{a}_z}{2\pi(a^2 + z_0^2)^{3/2}}$$
$$\mathbf{m} = I(\pi a^2) \mathbf{a}_z$$

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- Torque on a Large-Scale Current Loop

If we remove the restriction on differential size, **and assume *uniform magnetic flux density over the entire loop area***, the differential result

$$d\mathbf{T} = d\mathbf{m} \times \mathbf{B}$$

becomes:

$$\mathbf{T} = I\mathbf{S} \times \mathbf{B} = \mathbf{m} \times \mathbf{B}$$

These results are **independent of the shape of the loop** -- the area and orientation are all that matter.

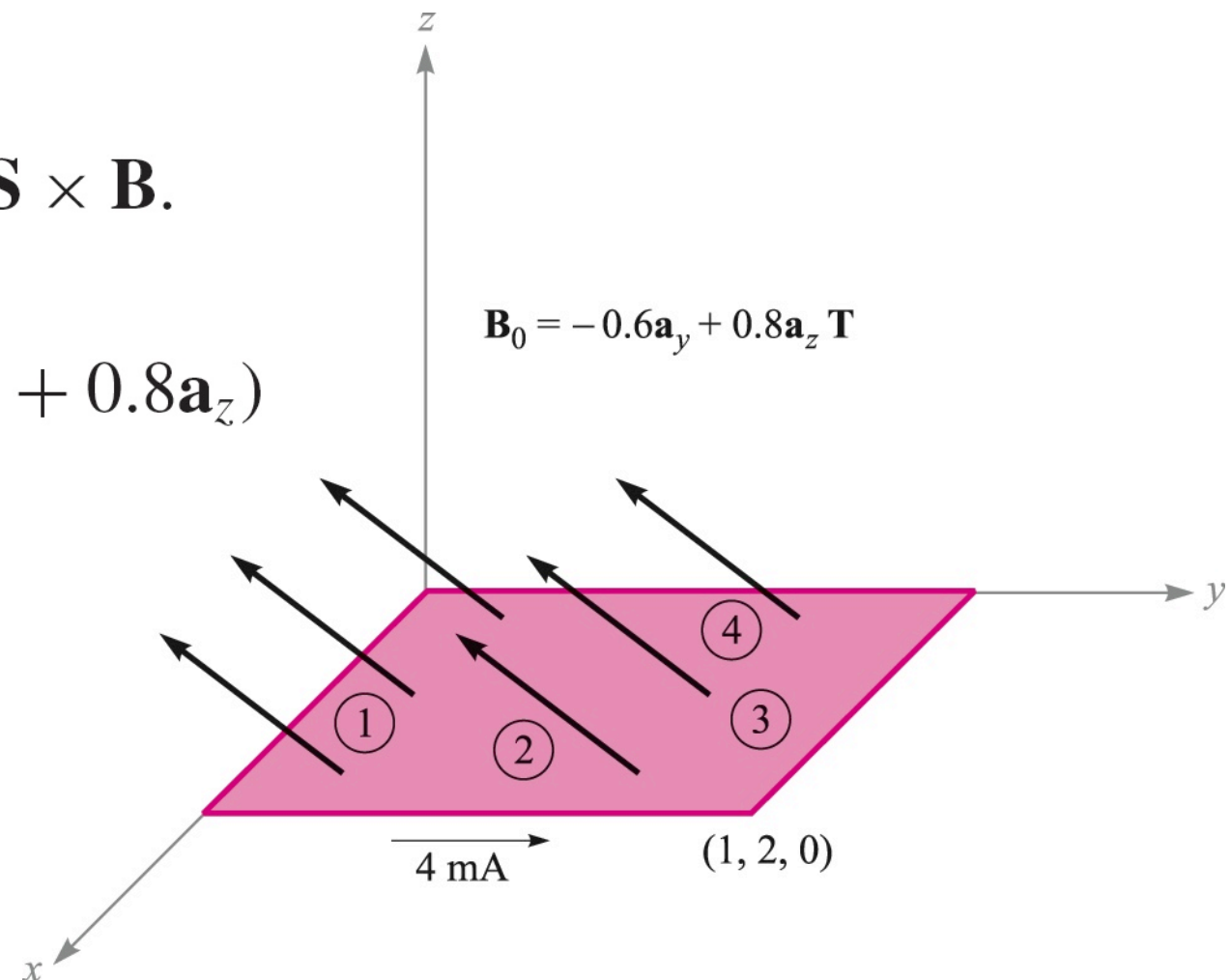
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- Example

Calculate the torque by using $\mathbf{T} = I\mathbf{S} \times \mathbf{B}$.

$$\begin{aligned}\mathbf{T} &= 4 \times 10^{-3} [(1)(2)\mathbf{a}_z] \times (-0.6\mathbf{a}_y + 0.8\mathbf{a}_z) \\ &= 4.8\mathbf{a}_x \text{ mN} \cdot \text{m}\end{aligned}$$



Thus, the loop tends to rotate about an axis parallel to the positive x axis. The small magnetic field produced by the 4 mA loop current tends to line up with $\mathbf{B}_0$.

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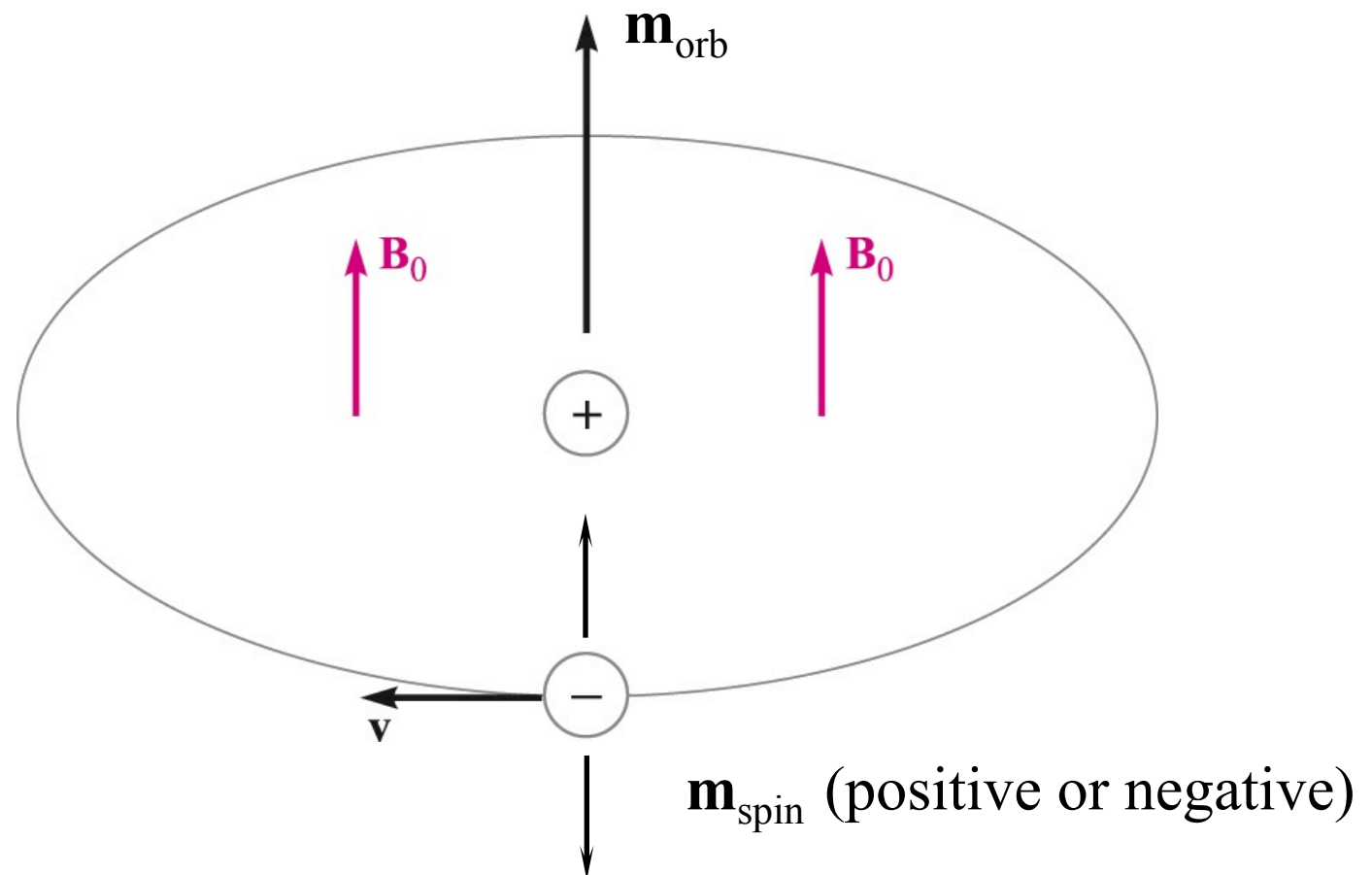


- Magnetic Materials

Materials are characterized according to their configurations of -- and interactions between -- atomic magnetic moments. **On the atomic or molecular scale, magnetic moments are associated with electron orbital configurations, electron spin自旋, and (to a lesser extent) nuclear spin.**

The diagram shows contributions to the total **magnetic moment** of an atom that arise from the electron orbit ($\mathbf{m}_{\text{orb}}$) and electron spin ($\mathbf{m}_{\text{spin}}$), in which the latter vector **may add to or subtract from** the orbital moment.

The magnetic flux density, $\mathbf{B}_0$, is externally applied.



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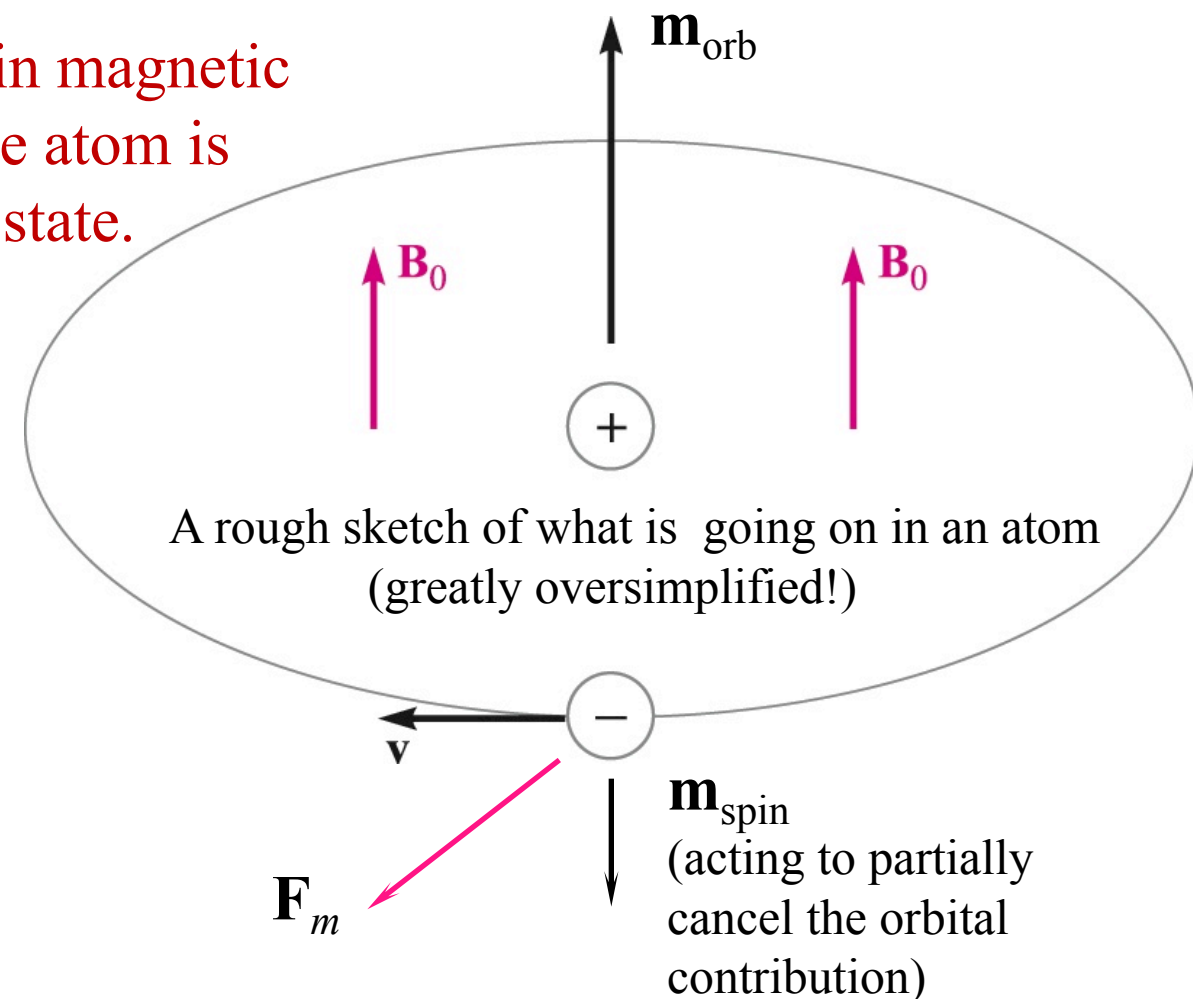
- Magnetic Materials: Diamagnetism 抗/反磁性

In this case, the summation of all orbital and spin magnetic moments associated with all electrons in a single atom is precisely zero, with the atom in its unperturbed state.

Application of a magnetic field leads to an outward radial force $\mathbf{F}_m$ on the orbiting electron shown here, resulting in a decrease in its orbital velocity in order to maintain a balance of Coulomb and magnetic forces.

The velocity reduction will result in a lessening of the orbital magnetic moment, thus producing a very small net magnetic moment in the direction of the spin moment; i.e., in the opposite direction to the applied field.

So a net decrease in the magnetic flux density will occur.



Examples of diamagnetic materials: metallic bismuth 铋, copper, gold, silicon, germanium 锗, graphite 石墨, sulfur 硫.

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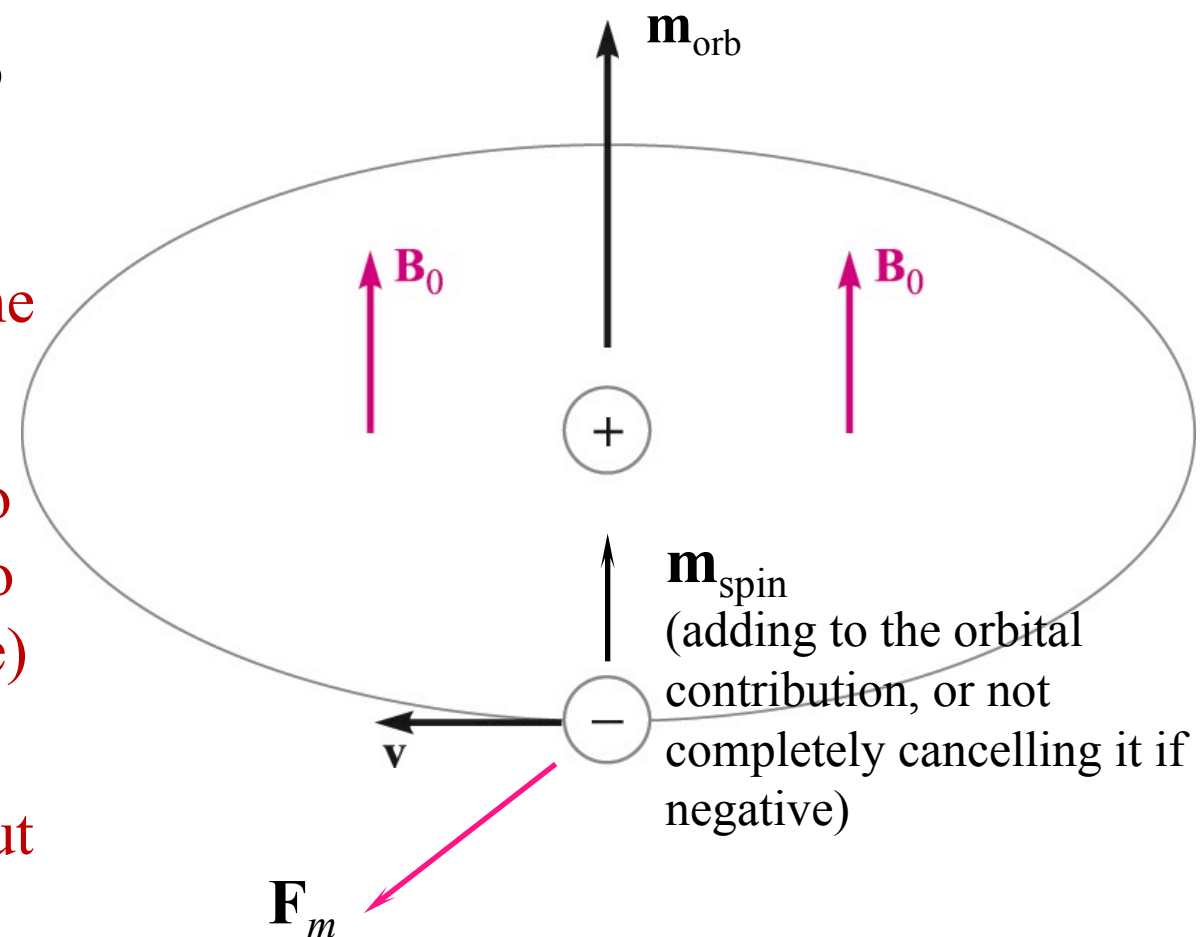
• Magnetic Materials: Paramagnetism 顺磁性

In this case, the summation of all orbital and spin magnetic moments associated with individual atoms is not zero, but is very small. Because of random orientations of adjacent atoms, however, the *average* magnetic moment in the material is zero.

Application of a magnetic field on the non-zero moments introduces a torque on the ensemble which results in the *partial alignment* of the moments of adjacent atoms. This strengthens the overall magnetic flux density.

However, the diamagnetic effect, in addition to the tendency toward random orientations due to molecular activity (increasing with temperature) reduces the net increase in $\mathbf{B}$.

So a net *increase* in the $\mathbf{B}$ field will be seen, but this increase is very small.



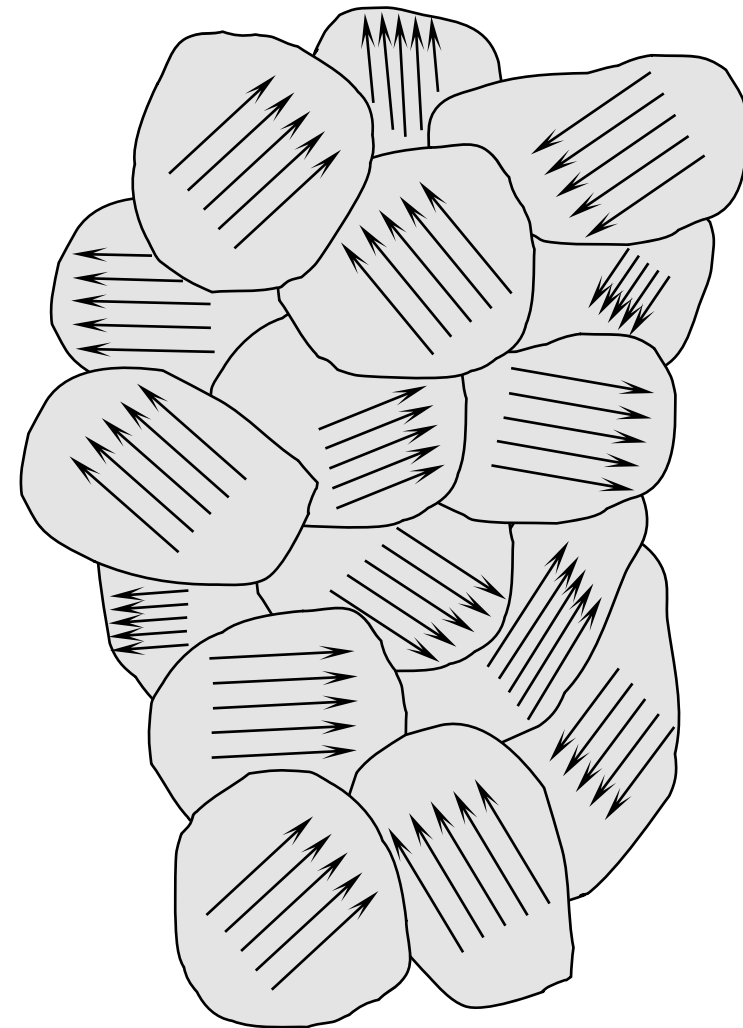
Examples of paramagnetic materials: potassium钾, tungsten钨, rare earth elements稀土

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- Magnetic Materials: Ferromagnetism 铁磁性

In ferromagnetic materials, **fairly strong atomic or molecular magnetic moments exist. Moments of adjacent molecules interact to cause partial alignment of moments within small regions known as *domains* 磁畴.** Magnetic moment orientations from domain to domain are random, thus resulting in a zero overall magnetic moment for the material.



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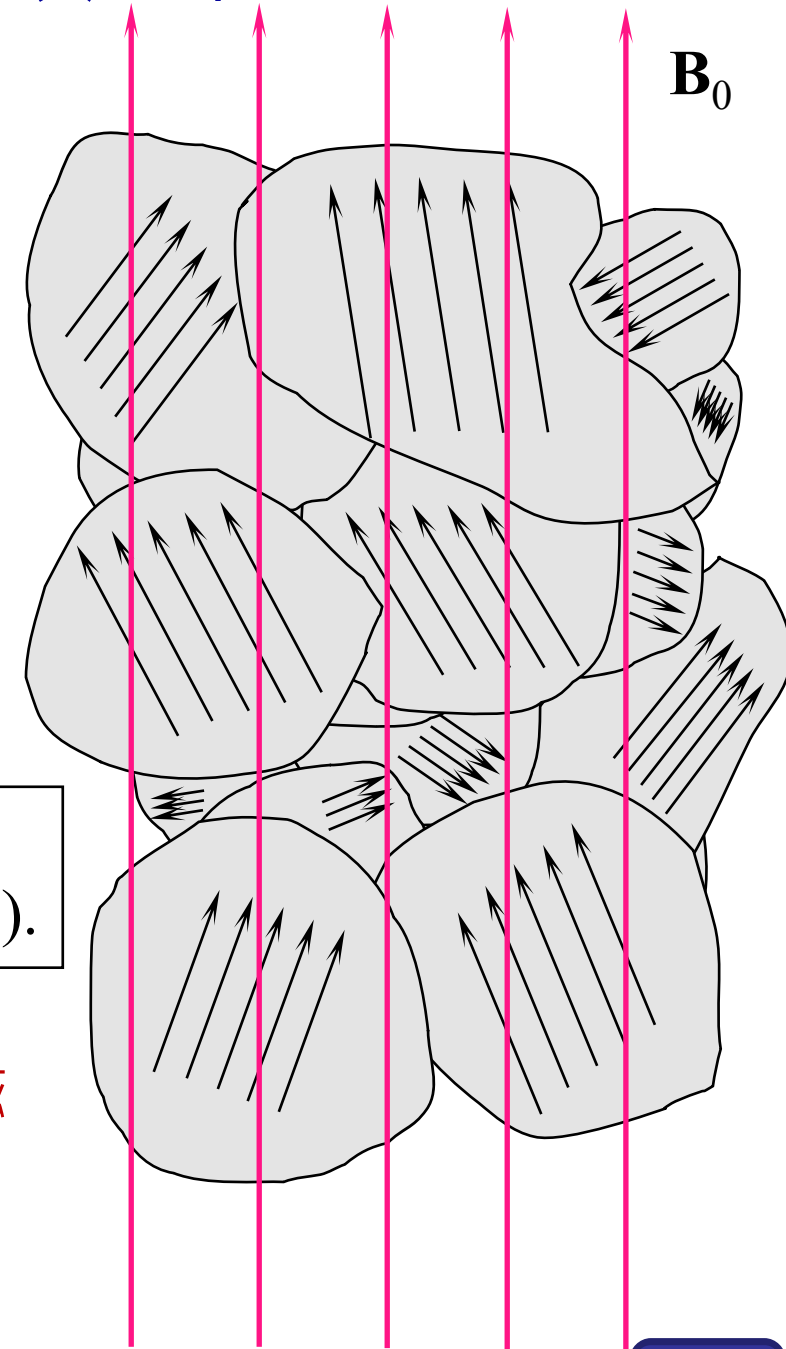
- Magnetic Materials: Ferromagnetism 铁磁性

Application of a steady magnetic field produces greater torque on magnetic moments that are closer to alignment with $\mathbf{B}_0$. Average alignment throughout the material thus increases, **but does so through the enlargement of the nearly-aligned domains, at the expense of those that are not aligned with $\mathbf{B}_0$.**

A significant increase in the overall magnetic flux density occurs.

Ferromagnetic elements: iron, nickel 镍, cobalt 钴 (room temperature), gadolinium 钆, dysprosium 镝 (low temperature).

Another manifestation is found in *superparamagnetic* 超顺磁性 **materials**, consisting of ferromagnetic particles suspended in a non-ferromagnetic matrix (magnetic recording tape is the most common example)



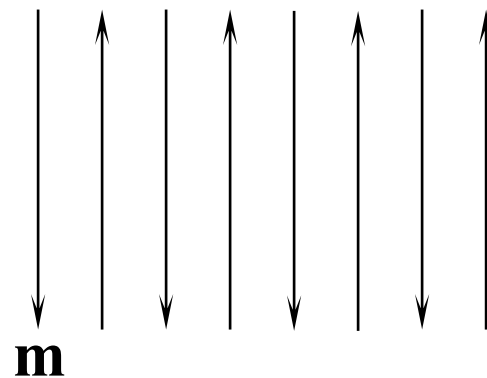
11: The Steady Magnetic Field



- Magnetic Materials: Antiferromagnetism 反铁磁性

In certain materials, and at low temperatures, adjacent magnetic moments are oriented in opposite directions, this being the lowest energy state for the ensemble.

The application of an external field produces no change in the net flux density .



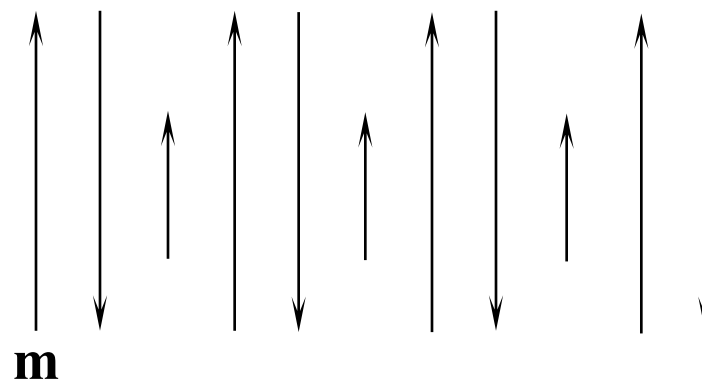
Example of antiferromagnetic materials: manganese oxide 氧化锰, nickel oxide, ferrous sulfide 硫化亚铁, cobalt chloride 氯化亚钴

11: The Steady Magnetic Field



- Magnetic Materials: Ferrimagnetism 亚铁磁性

In certain materials, the lowest energy configuration involves **adjacent magnetic moments that are oriented with opposing orientations, but with different magnitudes as shown**. The application of **an external field produces a significant change in the net flux density, but not as strong as in ferromagnetic materials**. The most important materials in this class are the *ferrites* 铁氧体, distinguished by having low conductivity 很低的电阻率.



Example of ferrimagnetic materials: iron oxide magnetite 氧化铁磁体 (Fe_3O_4) , nickel-zinc Ferrite 镍锌铁氧体, nickel ferrite 镍铁氧体.

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- Magnetic Material Summary

Classification	Magnetic Moments	B Values	Comments
Diamagnetic	$\mathbf{m}_{\text{orb}} + \mathbf{m}_{\text{spin}} = 0$	$B_{\text{int}} < B_{\text{appl}}$	$B_{\text{int}} \doteq B_{\text{appl}}$
Paramagnetic	$\mathbf{m}_{\text{orb}} + \mathbf{m}_{\text{spin}} = \text{small}$	$B_{\text{int}} > B_{\text{appl}}$	$B_{\text{int}} \doteq B_{\text{appl}}$
Ferromagnetic	$ \mathbf{m}_{\text{spin}} \gg \mathbf{m}_{\text{orb}} $	$B_{\text{int}} \gg B_{\text{appl}}$	Domains
Antiferromagnetic	$ \mathbf{m}_{\text{spin}} \gg \mathbf{m}_{\text{orb}} $	$B_{\text{int}} \doteq B_{\text{appl}}$	Adjacent moments oppose
Ferrimagnetic	$ \mathbf{m}_{\text{spin}} \gg \mathbf{m}_{\text{orb}} $	$B_{\text{int}} > B_{\text{appl}}$	Unequal adjacent moments oppose; low σ
Superparamagnetic	$ \mathbf{m}_{\text{spin}} \gg \mathbf{m}_{\text{orb}} $	$B_{\text{int}} > B_{\text{appl}}$	Nonmagnetic matrix; recording tapes

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- Magnetization 磁化, 磁化强度

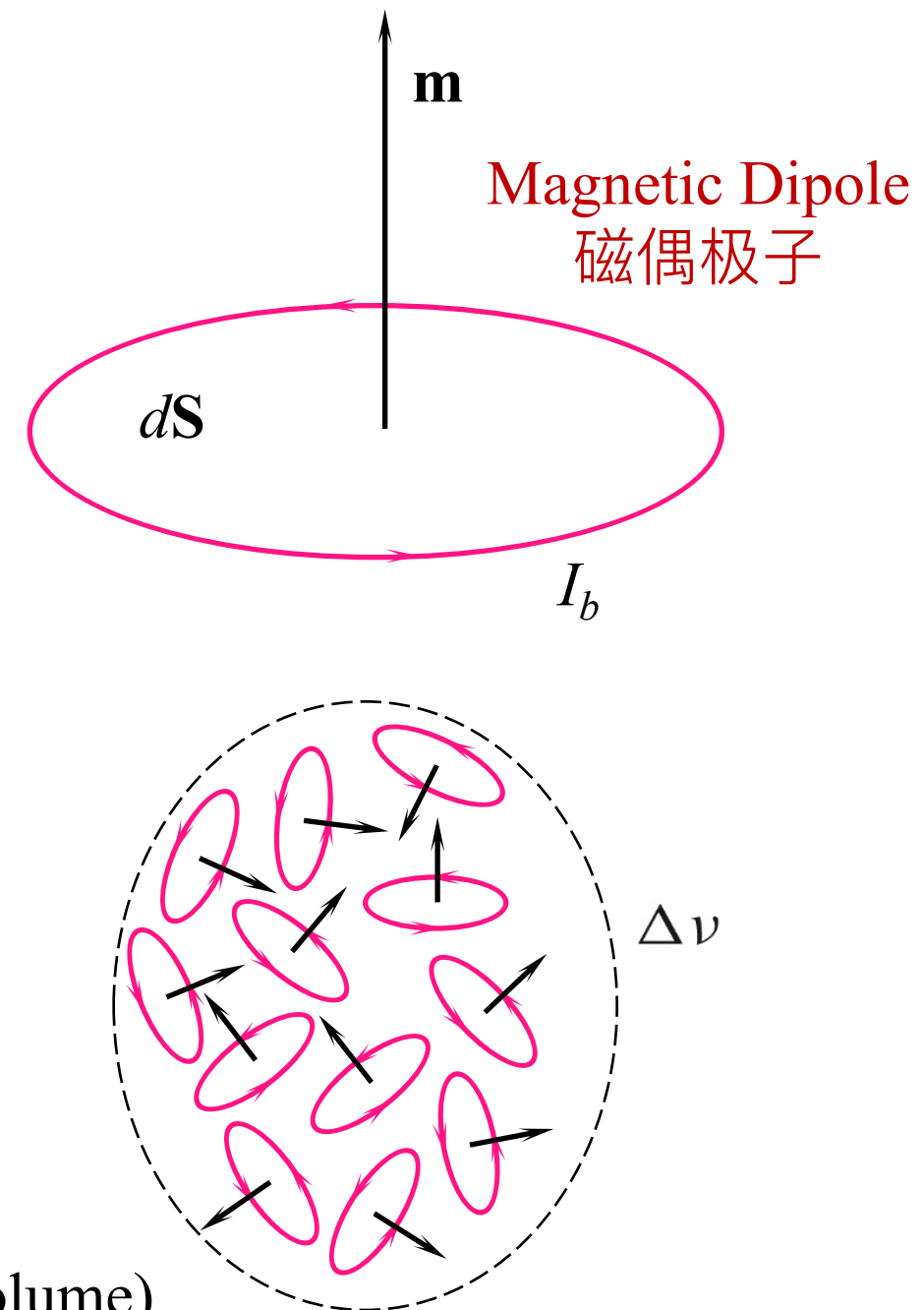
Consider a **bound current**, I_b , surrounding a differential area, dS . The magnetic moment is

$$\mathbf{m} = I_b d\mathbf{S}$$

Then, if **several such moments exist within a volume** Δv , then **the total magnetic moment** within that volume will be the vector sum of the contributors, or:

$$\mathbf{m}_{\text{total}} = \sum_{i=1}^{n \Delta v} \mathbf{m}_i$$

where n is the **dipole volume density** (dipoles per unit volume)



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- Magnetization 磁化强度

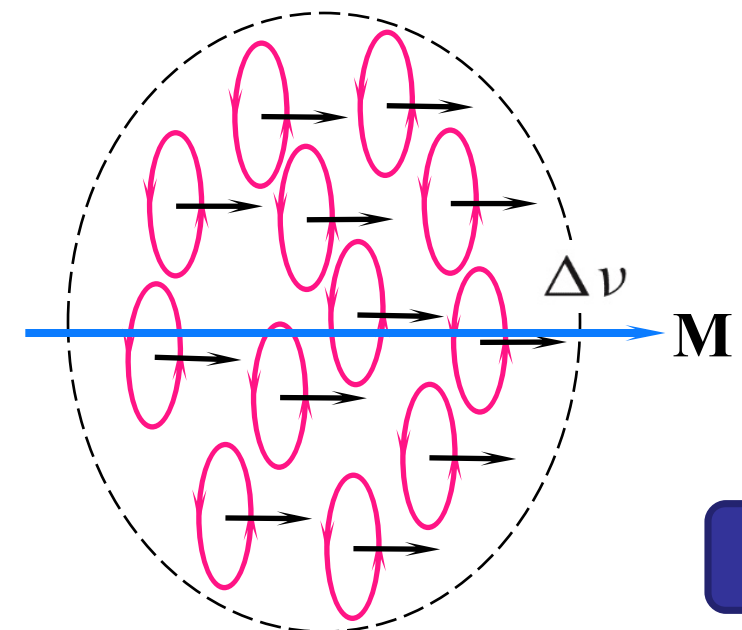
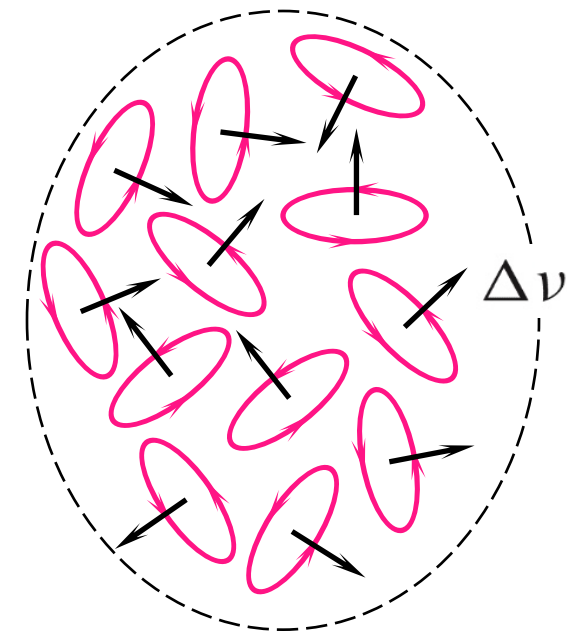
The *magnetization* 磁化强度 is the dipole moment per unit volume, in the limit as the volume shrinks to a point:

$$\mathbf{M} = \lim_{\Delta v \rightarrow 0} \frac{1}{\Delta v} \sum_{i=1}^{n \Delta v} \mathbf{m}_i$$

Units are dipole moment per unit volume, or Amps/m

If all dipoles are identical, each having moment $\mathbf{m}$, and if they all have the same orientation, then the magnetization simplifies to:

$$\mathbf{M} = n\mathbf{m} = nI_b d\mathbf{S}$$



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- Relation Between $\mathbf{B}$ and $\mathbf{H}$

$$\mathbf{B} = \mu_0(\mathbf{H} + \mathbf{M})$$

$$\mathbf{H} = \frac{\mathbf{B}}{\mu_0} - \mathbf{M}$$

Define: $\mathbf{M} = \chi_m \mathbf{H}$

where χ_m is magnetic susceptibility 磁化率

So now.. $\mathbf{B} = \mu_0(\mathbf{H} + \chi_m \mathbf{H}) = \mu_0 \mu_r \mathbf{H}$

where the *relative permeability* 相对磁导率 is:

$$\mu_r = 1 + \chi_m$$

...so that the *permeability* is now:

$$\mu = \mu_0 \mu_r$$

and consequently:

$$\mathbf{B} = \mu \mathbf{H}$$

